

1. Introduction

This project documents a basic Linux system security audit performed in an Ubuntu 24.04.4 LTS environment running through Windows Subsystem for Linux (WSL2). The audit includes system identification, account and identity checks, file and directory permissions, running-process inspection, system-log inspection, and a basic check for files with special SUID permissions.

2. Objectives

- Identify the Linux operating system, kernel, architecture, and hostname.
- Verify the current user and group information.
- Inspect important system accounts and the password-file entries.
- Review file and directory permissions and demonstrate permission changes.
- Inspect running processes and current system resource usage.
- Review available system logs under /var/log.
- Identify files having SUID permission bits for security review.

3. System Information

Item	Observed Information
Operating System	Ubuntu 24.04.4 LTS (Noble Numbat)
Kernel	Linux 6.18.33.2-microsoft-standard-WSL2
Architecture	x86-64
Virtualization	WSL
Hostname	DESKTOP-I984VC0
Python	Python 3.12.3

4. Commands and Audit Evidence

Command	Purpose / Result
hostname	Verified the system hostname.
hostnamectl	Displayed hostname, operating system, virtualization, kernel, and architecture information
uname -a	Displayed detailed kernel and system information.
cat /etc/os-release	Confirmed Ubuntu 24.04.4 LTS.
python3 --version	Confirmed Python 3.12.3.
whoami	Confirmed the current user as student.

Command	Purpose / Result
id	Displayed UID, GID, and group membership for the student account.
cat /etc/passwd	Reviewed local account entries and login shells.
ls -la	Inspected hidden files and permission information in the home directory.
chmod 755 project.txt / chmod 644 project.txt	Demonstrated changing file permissions and verified the resulting permission modes.
ps aux	Displayed currently running processes.
top	Displayed live process, CPU, memory, and system-load information.
ls /var/log	Listed available system and application log files.
find / -perm -4000 2>/dev/null	Searched for files with the SUID permission bit.

5. Security Observations

- The system was identified as Ubuntu 24.04.4 LTS running under WSL2.
- The current account was verified using `whoami` and `id`.
- The `/etc/passwd` file was inspected to review system and service accounts.
- File permissions were explicitly changed from 755 to 644 for `project.txt`, demonstrating permission control.
- Running processes were inspected using `ps aux` and `top`.
- The `/var/log` directory was inspected for available system and application logs.
- The SUID search returned standard system executables such as `su`, `sudo`, `mount`, `passwd` and related utilities. These should be reviewed according to their expected system role rather than automatically treated as vulnerabilities.

6. Basic Security Recommendations

- Use the principle of least privilege for user accounts and files.
- Keep Ubuntu packages updated with security updates.
- Review unusual or unnecessary user accounts and login shells.
- Regularly inspect important logs for unexpected authentication or system activity.
- Review SUID files and retain only those required by the operating system or installed software.
- Avoid using unnecessarily permissive file modes for sensitive files.
- Use strong authentication practices and avoid sharing passwords.

7. Conclusion

The Linux security audit successfully covered basic system identification, user and account inspection, file-permission verification, process monitoring, log-directory inspection, and SUID-file discovery. The collected command outputs provide evidence of the audit activities performed in the Ubuntu WSL2 environment. This report contains the written report content only; screenshots are intentionally not embedded.

Note: This report was prepared from the screenshots and command outputs supplied in the conversation. The original Project 2 instruction PDF was not available as readable text in the supplied file, so requirements that cannot be verified from that PDF are not claimed here.